Md Kaif Afran Khan

AI & Data Engineer | Researcher

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PROFESSIONAL SUMMARY

Versatile AI and Data Engineer with a strong foundation in computer science and a proven track record of building scalable data pipelines, production-grade AI applications, and conducting high-impact academic research. Experienced in leading global project teams, deploying cloud-based architectures (AWS), and integrating advanced machine learning models. Published IEEE author with ongoing research in Quantum Machine Learning and Large Language Model optimization.

EDUCATION

North South University *Dhaka, Bangladesh*

B.Sc. in Computer Science & Engineering *Expected 2026*

CGPA: 3.62/4.00 | Distinction: Cum Laude | 25% Merit Scholarship

Dhaka Residential Model College *Dhaka, Bangladesh*

Higher Secondary Certificate (HSC) & Secondary School Certificate (SSC)

PROFESSIONAL EXPERIENCE

AI Engineer *The Data Island* *2025 – Present*

- Architected and built an automated form-filling system for Aselo, extracting structured form data from S3 transcripts using AWS Bedrock (Claude).
- Engineered a scalable backend using FastAPI and AWS S3, with a responsive React + TypeScript frontend.
- Implemented MFA session support and real-time chat view, ensuring the architecture is serverless-ready (Lambda + API Gateway) or containerized for ECS deployment.

Project Manager Trainee *Excelerate* *Remote*

- Collaborated with a cross-functional team of members from different countries to plan a global 10-day AI event.
- Managed project documentation, allocated tasks, and scheduled meetings to ensure timely execution of deliverables.
- Utilized project management tools and communication platforms to coordinate across multiple time zones, enhancing global teamwork and leadership skills.

Core Volunteer & Tech Lead *Farzana Afroz Foundation (NGO)* *2022 – Present*

- Lead the technical development and maintenance of the organization's digital platforms and currency technology initiatives.
- Managed and scaled the foundation's official website (farhanaafrozfoundation.org), ensuring high availability and dynamic content management.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript/JavaScript, SQL, PHP, C++

Data Engineering: Apache Kafka, Apache Spark (Structured Streaming), dbt, Apache Airflow, Prefect, Medallion Architecture, ETL/ELT Pipelines

Cloud & DevOps: AWS (S3, Lambda, Glue, Athena, EventBridge, ECS/Fargate, EC2, Bedrock), Docker, Redis, CI/CD

AI & Machine Learning: LLMs (Prompt Engineering, LoRA Fine-tuning), RAG, AWS Bedrock (Claude), Scikit-learn, XGBoost, SHAP, OpenRouter
Quantum Machine Learning: Qiskit, PennyLane, JAX, Cirq, Quantum Neural Networks (QNNs), Variational Quantum Circuits (VQCs), QKD
Backend & Databases: FastAPI, Flask, Node.js, MongoDB, PostgreSQL, MySQL, Supabase, Sanity CMS
Frontend: React, Next.js, Tailwind, Bootstrap, React Native (Expo)

DATA ENGINEERING PROJECTS

MarketPlus – Cloud-Based Stock Analytics Pipeline
AWS S3, Athena, Glue, Spark, Prefect, FastAPI, React

- Engineered a robust Medallion Architecture (Bronze → Silver → Gold) data pipeline for processing stock market data.
- Implemented cloud ingestion using AWS EventBridge and Lambda (Alpha Vantage API) with local Spark processing orchestrated by Prefect.
- Computed advanced analytics (daily returns, moving averages, volatility, top movers) and exposed data via FastAPI for a React dashboard.

CryptoStream – Real-Time Market Data Pipeline
Kafka, Spark Structured Streaming, Airflow, dbt, Postgres, Docker

- Built an end-to-end real-time streaming pipeline ingesting WebSocket data through Kafka and Spark Structured Streaming into Postgres.
- Utilized dbt for Silver and Gold transformations, orchestrated by Airflow (running every 5 minutes).
- Generated 1-minute OHLCV candles and moving averages, complete with health monitoring and a full Docker Compose stack.

AI & SOFTWARE ENGINEERING PROJECTS

STORE – Multi-Tenant Smart ERP Platform
FastAPI, MongoDB, React, React Native, Redis, Docker

- Developed a full ERP platform with three surfaces: Backend API, Web Dashboard, and Mobile Worker App.
- Integrated an AI Assistant (OpenRouter) and enforced multi-tenancy via `businessId`. Modules include Auth/RBAC, Inventory, Sales, and Analytics.

Denz E-Commerce (Live: denzapparel.com)
Next.js 14, TypeScript, Supabase, Tailwind

- Built a modern full-stack e-commerce platform featuring Google OAuth, an admin dashboard, and secure role-based access.
- Implemented HttpOnly cookies, Row Level Security (RLS), and Cloudinary integration for media management.

PUBLICATIONS

Explainable Hybrid ARIMA–ML Forecasting of Bangladesh's GDP: Sectoral Energy Consumption and Carbon Emission Dynamics
5th IEEE International Conference on Robotics, Automation, Artificial-intelligence and Internet-of-Things (2026)

- Developed a Hybrid Residual Learning Ensemble (Hybrid-Stack): ARIMA + ensemble of XGBoost, Random Forest, GBM, SVR on residuals.
- Implemented SHAP explainability. Achieved Best MAPE = 7.60%, outperforming ARIMA and standalone ML baselines.

FSM-LoRA: Dynamic Gradient Gating via Finite State Machines for Energy-Efficient LoRA Fine-Tuning
2nd International Conference on AI and Emerging Technology for Sustainable Future (2026)

- Designed a Finite State Machine controller that dynamically skips redundant backward passes during LoRA fine-tuning.
- Skipped 15–58% of backward passes, achieving up to 58.58% wall-clock time reduction with accuracy within ±0.5% of standard LoRA.
- Evaluated on Phi-2, Qwen2.5-3B, and OpenLLaMA-3B models.

MANUSCRIPTS UNDER REVIEW

Benchmarking Large Language Models Against Classical Machine Learning for Undergraduate Academic Performance Prediction
29th International Conference on Computer and Information Technology

- Compared 7 classical ML models vs. 5 LLMs (zero-shot, few-shot, LoRA) on 872 student records.
- Best ML (Extra Trees): 88.57% accuracy. Best fine-tuned LLM (Gemma-2-9B-IT): 56.57% accuracy.
- Conclusion: Classical ML significantly outperforms current LLMs on structured academic prediction tasks.

TAL-GRN: Topology-Aware Learning Graph Representation Network for Long-Tailed Multi-Class Medical Image Classification

- Utilized adaptive k and learned edge prediction/pruning to enhance F1-score on imbalanced multi-class medical image datasets.

Persona Multiplicity and Deliberative Synthesis in a Single Large Language Model: A Stochastic Multi-Agent Study of Moral Game-Theoretic Choice

- Developed a multi-persona moral reasoning framework generating dynamic personas with independent reasoning and aggregation.
- Outperforms single-shot baselines in stakeholder and consequence coverage metrics.

ONGOING RESEARCH

Quantum Machine Learning for Quantum Key Distribution (QML as QKD)

- Investigating noise types and innovations in BBM92 and E91 quantum key distribution protocols using QML frameworks.
- Exploring entanglement pathways and their impact on secure communication protocols.

Piqagat: A Quantum Adaptive GAT Framework for Muon Regression

- Developing a Quantum Adaptive Graph Attention Network (GAT) framework for muon regression tasks.
- Integrating quantum circuit architectures with classical graph attention mechanisms to enhance predictive performance.

CERTIFICATIONS & AWARDS

Certifications: AWS Data Engineer Practitioner (Coursera), Full Stack Developer (MERN Stack - Ostaad), Data Science Short Course (Stanford Online - Jennifer Widom).
Awards: 25% Merit Scholarship (NSU), Cum Laude Distinction (NSU).
Languages: English (IELTS 7.5), Bangla (Native).